HHS Public Access
Author manuscript
Nat Rev Cancer. Author manuscript; available in PMC 2023 April 01.

Published in final edited form as:
Nat Rev Cancer. 2022 May ; 22(5): 253–254. doi:10.1038/s41568-022-00458-1.

Expanding oesophageal cancer research and care in eastern Africa

The African Esophageal Cancer Consortium*

The African Esophageal Cancer Consortium is a self-organized oesophageal cancer research consortium of more than 80 physicians and scientists working at ten sites in nine countries of eastern and southern Africa. We study the aetiology of this highly fatal cancer and are expanding the clinical capacity to improve cancer care.

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Competing interests
The authors declare no competing interests.

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Oesophageal cancer is the 7th most common cancer and the 6th most common cause of cancer death worldwide. Its burden varies globally, with high-risk belts across central Asia and down the eastern side of Africa, and in these belts, 90% of the cases are oesophageal squamous cell carcinoma (ESCC)[1]. The African high-risk belt largely follows the Rift Valley but extends southward to the Eastern Cape Province of South Africa. Many patients present with advanced-stage disease, with severe oesophageal obstruction, and this, combined with limited treatment resources, typically results in survival of less than 6 months. ESCC is one of the most common fatal cancers in both sexes in most of these high incidence countries, but historically there has been little etiologic or clinical research or advocacy devoted to this disease, and few countries include it in their cancer control programmes. A unique feature of Africa's ESCC belt is the young age of many patients. Most series report a mean age of patients in the 50s, with up to 18% of patients presenting before age 40, and a few cases presenting before 20 years of age[2]. Such a young age distribution of cases is not seen anywhere else in the world, and it is not explained by the age distribution of the general population.

In 2017, African and international researchers established the African Esophageal Cancer Consortium (AfrECC), with the goal of reducing the burden of ESCC by coordinating aetiologic and clinical studies, increasing African research and clinical capacity, and increasing awareness of ESCC among affected populations, physicians, policy-makers and funders[3]. To further our understanding of the causes of ESCC in this region, AfrECC members have conducted seven case–control studies across Kenya, Zambia, Tanzania and Malawi, together enrolling over 2,000 cases and 2,000 controls so far. So far, risk factors reported by one or more sites include exceptionally high polycyclic aromatic hydrocarbon (PAH) exposure, especially in women, most likely owing to indoor cooking with biomass fuel (usually wood) in poorly ventilated kitchens[4], as well as increased risk associated with poor oral health and hygiene[5], tobacco and alcohol use, drinking hot beverages and low socioeconomic status[6]. AfrECC is also conducting a collaborative ESCC genome-wide association study (GWAS) in eastern African countries, which will complement another ESCC GWAS in South Africa.

Important etiological questions remain. Why are there many young patients with ESCC in this region? What is driving the development of ESCC in alcohol and tobacco abstainers? What explains the striking geographic differences in ESCC risk between eastern and western sub-Saharan Africa, and even important differences within the high-risk countries?

As untreated patients with ESCC often have severe dysphagia and die from malnutrition and dehydration, increasing access to palliative, self-expanding metal stents (SEMS) has been a clinical priority for the consortium. This AfrECC Stent Access Initiative, launched in partnership with Boston Scientific Corp. (Marlborough, MA), provides subsidized SEMS and training of local endoscopists on safe stent use[7]. Also underway is a multicentre prospective patient cohort study investigating the comparative effectiveness of SEMS and other palliative interventions for advanced ESCC.

Another clinical priority for AfrECC is secondary prevention of ESCC, which will likely involve screening for oesophageal squamous dysplasia (ESD), the asymptomatic precursor lesion. An AfrECC endoscopic screening study of 305 asymptomatic adults in Kenya found a 3% prevalence of moderate to severe ESD[8], an important target for endoscopic treatment. Another study showed limited endoscopy capacity in eastern Africa, with one endoscopist per 400,000–2,000,000 people and only two functioning gastroscopes per facility[9], limiting the current possibilities for wide-spread endoscopic screening. A third study of 102 asymptomatic volunteers in Tanzania showed the safety, acceptability and feasibility of low-cost, minimally invasive oesophageal cell collection using encapsulated sponges[10].

In all activities, the consortium has emphasized local leadership and capacity building, including training of study leaders, epidemiologists, field staff, bio-specimen handlers, data managers, bio-statisticians, endoscopists, nurses and technicians. An important partner in clinical training has been the American Society for Gastrointestinal Endoscopy, which has provided AfrECC sites with video and on-site training. Also, a promising recent pilot programme used high fidelity real-time links to connect African endoscopy suites with both African and American training sites.

Our experience to date has confirmed the importance of diverse partnerships in developing the AfrECC Regional Network. On the local level, sustained interest by hospital leadership, senior staff and trainees is necessary to initiate and complete etiologic and clinical research studies, and a functioning endoscopy unit with well-trained staff is needed to achieve meaningful clinical training, improved endoscopy services and effective endoscopic palliation. On the national level, sustained government engagement is required to allow registration and importation of equipment and supplies, to increase population-level awareness of oesophageal cancer, and to include this cancer in national cancer control programmes. At the international level, exchanges, training and funding provided by institutions with strong global health research programmes have been instrumental in supporting AfrECC etiologic and clinical studies. The importance of private industry partners is demonstrated by the AfrECC Stent Access Initiative, and partnering with professional societies is crucial for expanding clinical capacity.

AfrECC has made significant progress in the past four years by developing a collaborative organization in which African and international partners are working together to reduce the burden of ESCC on the continent. However, major challenges remain. First, global funding for oesophageal cancer is much less than that for prostate, pancreatic or cervical cancer, tumours that kill similar numbers of people worldwide, and most global oesophageal cancer funding goes to oesophageal adenocarcinoma research. Potentially modifiable risk factors, early detection strategies and genomic analysis of tumours require further investigation, but these cannot be evaluated without additional research funding devoted to ESCC. Of particular need is funding for large prospective, longitudinal cohort studies with cancer outcomes, which can best explore cancer aetiology. Second, there is a great need across the ESCC high-risk corridor for national and regional advocacy groups focused on increasing awareness of ESCC among populations, clinicians, policy-makers and funders. Third, as noted above, endoscopic capacity is severely limited throughout eastern and southern Africa, which affects disease prevention, diagnosis and therapy. A significant underlying problem is the cost of endoscopic equipment and its repair. There is a large potential market for solutions that meet the needs of limited-resource contexts, for example disposable endoscopes with no need for repair[9]. Fourth, despite the success of the AfrECC Stent Access Initiative in expanding access to palliative stents, to achieve acceptable patient survival, improved access to proficient surgical oncology care, systemic therapies, radiation therapy and palliative care is also needed. Fifth, AfrECC has demonstrated that international collaborations can lead to collective successes. However, for AfrECC and other similar disease consortia to be sustainable, there is a need for funding to support their infrastructure, which is not covered by research grants.

Many African people continue to suffer and die from ESCC, a disease that is likely to be preventable. AfrECC carries out research and programmes for primary prevention, early detection, treatment and palliation using context-appropriate technologies that are needed to make death from ESCC no longer common in Africa.

Acknowledgements

This work was supported in part by the Intramural Research Program of the National Cancer Institute, NIH.

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